
CAPSTONE PROJECT FITNESS PLANNER AI – PERSONALIZED WORKOUT & DIET PLANNER

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OUTLINE

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PROBLEM STATEMENT

- Personalized Workout & Diet Planner with AI
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- Most fitness apps give generic workout or diet plans that don't consider individual student needs, cultural food habits, or available resources. Students require a system that generates personalized routines and meal plans using AI, ensuring they are practical, budget-friendly, and effective.

PROPOSED SOLUTION

- AI-powered customization calculates BMR, TDEE, BMI, and calorie budgets in real-time.
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- 100% local processing runs the rules engine entirely on the backend server, protecting user privacy.
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- Scientific accuracy using Mifflin-St Jeor formulas and balanced macro planning.
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- Cultural diet customization tailored to North/South Indian, Jain, Vegan, and Mixed food preferences.
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- Free and open-source platform, removing financial barriers to personal coaching.

SYSTEM APPROACH

Frontend Tech Stack:

HTML5, Custom CSS3, and Modular Vanilla JavaScript for a fast, responsive Single Page Application (SPA).

Backend Server & Logic:

Node.js and Express.js REST API with a local rules engine running calculations.

Database & Security:

better-sqlite3 database for user profiles and datasets, secured with JWT and bcryptjs.

ALGORITHM & DEPLOYMENT

- BMR Calculation: Mifflin-St Jeor Formula $\hat{+}$ BMR = $(10 * W) + (6.25 * H) - (5 * A) + s$ (s is +5 for men, -161 for women).
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- TDEE Calculation: BMR * Activity Multiplier (1.2 for sedentary up to 1.9 for active).
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- Goal Matching: Muscle gain (+300 to +500 kcal) or fat loss (-500 to -750 kcal) adjust target intake.
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- Macro Distribution: Set to 1.6g/kg minimum protein, 50% carbohydrates, and 25% fats.
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- Deployment: CI/CD workflow deploying the Express server and client files to Vercel.

RESULT

- Fully functional, responsive web application launched and publicly accessible.
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- Instant generation of custom 7-day workout plans matching fitness level and available equipment.
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- Daily meal plans generated according to calorie budgets and macro goals.
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- Interactive user dashboard showing logs, stats, and a daily completion checklist.
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- Secure login sessions with profile editing and persistence.
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- Administrator portal to monitor user metrics and server statistics.

CONCLUSION

- Created a local, privacy-first alternative to commercial health and fitness tracking apps.
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- Proved that high-quality, scientifically sound personalization can be achieved locally without external AI APIs.
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- Removed the cost barriers of structured fitness coaching by offering a free platform.
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- Improved diet adherence by directly integrating cultural food profiles into the meal generator.

FUTURE SCOPE

- Mobile Applications: Port the responsive client to iOS and Android versions using React Native or Flutter.
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- Wearable Integration: Sync with smartwatches and health apps (Apple Health, Google Fit) for live tracking.
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- Conversational AI Coach: Integrate a lightweight local LLM (e.g., Llama 3) for conversational advice.
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- Progress Analytics: Add interactive graphs to track weight trends and strength progress over time.
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- Expanded Food Database: Include local grocery pricing and wider global cultural diets.

REFERENCES

Mifflin, M. D., et al. (1990). A new predictive equation for resting energy expenditure. AJCN.

Node.js Runtime & Express.js Web Application Framework Documentation.

JSON Web Token (JWT) Specifications & bcryptjs Encryption Standards.

pptxgenjs Presentation Library & SQLite Relational Database Engine.

APPLICATION LINKS

- Live Deployed Application:
- <https://fitness-seven-sand.vercel.app>
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- Source Code Repository:
- <https://github.com/adityadhanrajpathak/fitness-planner-ai>



THANK YOU